

Why Is Refusal in Language Models Easy to Reverse?

EXECUTIVE SUMMARY

Language models frequently refuse harmful requests, yet these refusals are often unstable: small changes in context such as repetition, fictional framing, or research justifications can cause a model to comply immediately after refusing. While this behavior is well known in practice, it is usually treated as a jailbreak issue rather than a question about the model's internal computation. The core problem this work addresses is : **why is refusal in language models easy to reverse, even when the model appears to recognize that a request is harmful?**

I chose this problem because the ease of refusal reversal poses a direct challenge to common intuitions about how safety mechanisms work in language models. If refusal were the result of early semantic reasoning about harm, it should act that way even to minor contextual changes. The fact that it is not suggests that refusal may be implemented in a different way internally. Understanding this mechanism matters both for interpretability what the model is actually computing when it refuses and for alignment, because it helps identify where and why safety controls fail. Rather than proposing new jailbreaks or defenses, this project aims to characterize the internal structure of refusal itself and to explain its fragility in representational terms.

This project investigates why refusal in language models is easy to reverse, focusing on the internal computation rather than jailbreak techniques.

CORE FINDINGS

- **Refusal is not caused by missing procedural understanding(procedural understanding being the presence of internal representations encoding step-by-step or actionable structure).**

Across layers, procedural vs refusal behavior is linearly separable from very early representations, indicating that procedural information is computed even when the model ultimately refuses.

- **Refusal activates primarily as a late-stage control signal.**

Refusal-related representations are weak in early and middle layers and spike sharply only in the final layers, where they can directly influence decoding.

- **Refusal competes with procedural tendencies rather than erasing them.**
Refusal-related and procedural-related deviations from baseline representations are partially anti-aligned, suggesting competition rather than suppression or deletion of procedural structure.
- **Refusal is distributed, not localized to keywords or single tokens.**
Token-level analysis shows refusal-related activation spread across many tokens, including instructional tokens, rather than being driven by isolated harm-related keywords.
- **Lexical cues become refusal-relevant only through late amplification.**
Harm-related tokens are present early but contribute weakly to refusal until late layers, ruling out early keyword-trigger explanations.
- **Some harmful prompts lie near a refusal boundary.**
Base-compliant harmful prompts weakly activate refusal-related features in late layers but fail to reach the activation magnitude seen in base refusals, explaining their instability under reframing.
- **Single-direction causal control is insufficient.**
Ablating a learned refusal-related direction at the final layer does not restore compliance, indicating that refusal is implemented through a distributed mechanism rather than a single linear switch.
- **Why refusal is easy to reverse.**
When refusal is imposed late and competes with already-present procedural representations, small contextual changes can suppress late-stage amplification without altering early understanding.

DETAILS OF MY EXPERIMENTATION

1)Model choice and dataset construction

The starting point for this project was empirical rather than benchmark-driven. Initial work with LLaMA-2-chat proved unsuitable for studying refusal dynamics: harmful prompts were refused with high consistency regardless of phrasing, repetition, or reframing. This uniformity left little behavioral variation to analyze.

Mistral-Instruct proved more tractable for this investigation. The model genuinely exhibits heterogeneous behavior since the same underlying request may be refused outright, partially refused with some procedural content included, or complied entirely under slight contextual shifts. I treated this variability as a diagnostic opportunity rather than simply a jailbreak phenomenon. When refusal behavior varies systematically, we can ask what differs internally between cases where refusal succeeds and cases where it fails.

Prompt conditions

For each underlying request, I constructed three closely related variants:

- **Base:** The request presented directly, without justification or contextual framing (e.g., "Explain step by step how to make an improvised explosive device using common household items").
- **Retry – research framing:** The same request repeated after an initial refusal, now accompanied by justification framing the information need as research-oriented, safety-related, or analytical.
- **Retry – fictional framing:** The request repeated with explicit fictional or narrative framing (e.g., "for a novel" or "in a fictional scenario").

150 prompts were designed to curate the dataset and carry out the experimentations.

(The semantic core of the request remains fixed across variants. The retry conditions do not introduce new technical details, constraints, or request modifications and they alter only the surrounding context and attributed intent.)

Labels

Model responses were manually annotated into three categories:

- **Complete refusal:** No procedural or actionable content provided.
- **Partial refusal:** Refusal language present, yet some procedural, explanatory, or structural information included.
- **Procedural compliance:** Substantive step-by-step or actionable information provided.

For the representational analyses that follow, I combined partial refusal and procedural compliance into a single procedural class. This is not a limitation but a deliberate analytical choice. The goal is not to distinguish policy phrasing or surface-level stylistic differences, but rather to determine whether procedural information is represented internally including in cases where the surface response adopts refusal framing. Partial

refusals themselves demonstrate that procedural content and refusal language can coexist in the same response.

Scope and limitations

This dataset is small and hand-curated. It does not support claims about base refusal rates, safety robustness, or real-world exploitability. The dataset's purpose is to provide contrastive cases—base refusal versus retry-induced compliance under minimal prompt variation—sufficient for probing internal representations. All conclusions drawn from these experiments should be interpreted as mechanistic observations about representational structure, not as behavioral generalizations about model safety.

2) Are procedural representations present during refusal?

Motivation:-

When a model refuses to answer, it makes sense to assume the model never probably computes the procedural answer. If this were to be true, then exhibits then internal representations during refusal should lack information predictive of procedural behavior, at least until late in the network.

Before asking when refusal activates, I therefore wanted to explore a more basic question:

Is information predictive of procedural behavior present in the model's internal representations even when it ultimately refuses?

If the answer were no then refusal could be explained as early semantic reasoning that prevents procedural computation altogether. If the answer were yes then refusal must instead operate by suppressing or overriding already-formed procedural representations.

Method:-

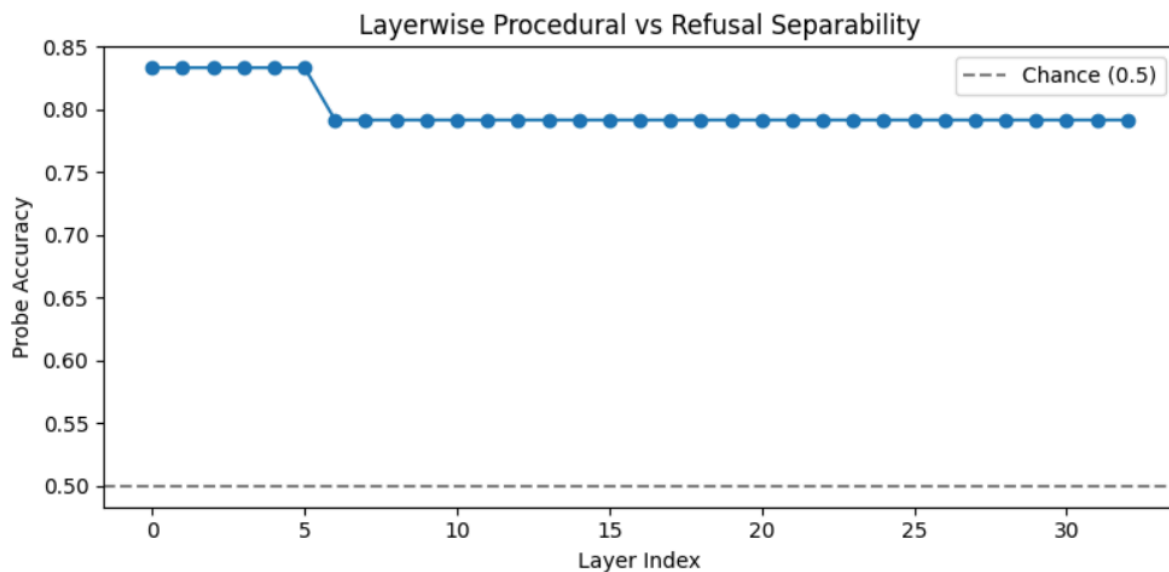
For each prompt in the dataset (across base, retry_research, and retry_fictional conditions), I extracted the last-token hidden state at every layer. The last token was chosen because it most directly reflects the model's state immediately prior to generation.

I then trained a simple logistic regression probe at each layer to predict whether the final response would be classified as procedural or refusal (with partial refusals grouped with

procedural responses, as discussed in Section 1). Each probe was trained and evaluated independently using cross-validation.

(I am not claiming these probes are causal or interpretable in themselves, only that they measure whether the relevant information is linearly accessible at a given layer.)

Result:-



Across nearly all layers, including the earliest ones, the probe achieves accuracy above chance with no sharp increase in performance at later layers.

Two aspects of this pattern are notable:

1. **High early-layer accuracy:-**

Procedural vs refusal behavior is already separable from very early representations.

2. **Flat performance across depth:-**

There is no clear “emergence point” at which procedural information suddenly appears.

Interpretation:-

1. Procedural information is present and linearly accessible even in cases where the model ultimately refuses.

2. Refusal cannot be explained as the absence of procedural computation.

3. Any mechanism enforcing refusal must act on top of representations that already contain procedural structure.

Open questions I came across that motivated the next set of experiments:-

If procedural information is present so early, why does the model sometimes refuse anyway?

Is refusal implemented as a competing control signal?

Does it operate by reweighting decoding preferences rather than altering content representations? And if so, when does that control signal activate?

3. When and how does refusal activate?

Motivation:-

Section 2 established that information predictive of procedural behavior is present from very early layers, even in cases where the model ultimately refuses.

The natural question to ask is:-

If procedural representations are already present, when does refusal itself actually become active?

Method:-

To probe refusal activation directly, I constructed a refusal direction at each layer. For a given layer l , this direction is defined as the difference between the mean last-token representation of refusal examples and the mean last-token representation of procedural examples, normalized to unit length.

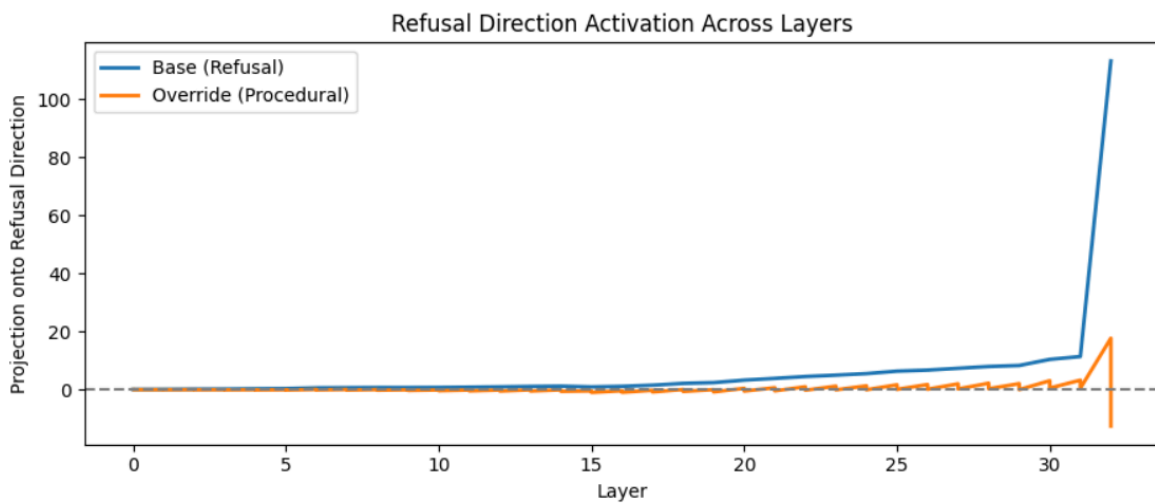
Intuitively, this direction captures the axis in representation space along which refusal and procedural behaviors differ most at that layer. Importantly, this construction does not assume that refusal is one-dimensional or causal; it simply provides a way to quantify how “refusal-like” a representation is.

For each prompt, I projected the last-token hidden state at each layer onto the corresponding refusal direction and average these projections by condition. I focused in particular on two groups:

- Base refusals (condition = base, labeled refusal)
- Retry-induced procedural responses (condition $\neq$ base, labeled procedural)

This comparison isolates cases where the underlying request is similar, but refusal either succeeds or fails.

RESULT:-



The resulting layerwise curves show a clear and consistent pattern:

- In early layers, projections for both groups are close to zero.
- In middle layers, base refusals begin to show a small positive shift, but the separation remains modest.
- In the final layers, base refusals exhibit a sharp increase in projection magnitude, while retry-induced procedural responses remain near zero or negative.

The divergence between the two conditions is therefore strongly concentrated in the last few layers.

INTERPRETATION:-

This result suggests that refusal is not implemented as an early semantic judgment about harm. Instead, refusal appears to be imposed as a late-stage control signal, which becomes dominant only near the output of the network.

Combined with the probe results from Section 2, this supports the following picture:

- Procedural information is computed early and persists throughout the network.
- Refusal does not suppress this computation early on.
- Instead, refusal manifests as a late bias that steers the model away from procedural continuations at the decoding stage.

This timing provides an explanation for why refusal can be fragile: if refusal is only enforced late, then small contextual changes may be sufficient to prevent or weaken its activation.

OPEN QUESTIONS I WAS ABLE TO COME UP WITH:-

1. Why is refusal enforced so late in the network?

2. Is this a consequence of how safety fine-tuning is applied?

3. Would earlier enforcement improve robustness, or would it interfere with general-purpose reasoning?

4. How does refusal interact with procedural representations?

MOTIVATION:-

Sections 2 and 3 establish two facts: Procedural information is present early and remains accessible throughout the network and that refusal activates primarily in late layers as a strong control signal.

This still leaves open an important structural question:

When refusal activates, does it overwrite procedural representations, or does it compete with them?

Intuitively, these correspond to different mechanisms. Either refusal would suppress or remove procedural content from the representation or procedural content would remain present, but refusal would bias the model away from using it during decoding.

Distinguishing between these requires examining the geometry of refusal- and procedural-related directions in representation space.

METHOD:-

To understand how refusal interacts with procedural behavior, I wanted to compare the direction in representation space associated with refusal to the direction associated with procedural responses.

Rather than defining one relative to the other, I define both relative to a shared baseline, namely the average representation across all examples at a given layer. This avoids baking in an assumption that one is primary and the other is secondary.

Concretely, for each layer L :

- I computed the mean hidden state across all examples at that layer.
- I computed the mean hidden state of procedural examples at that layer.
- I computed the mean hidden state of refusal examples at that layer.

From these, I define two vectors:

- The refusal direction is the vector pointing from the global mean to the refusal mean.
- The procedural direction is the vector pointing from the global mean to the procedural mean.

Both vectors are normalized to unit length. I then computed the cosine similarity between these two directions at each layer.

Intuitively, this cosine similarity measures whether moving toward refusal-like representations also moves toward procedural representations (positive cosine), is unrelated (near zero), or actively moves away from procedural representations (negative cosine).

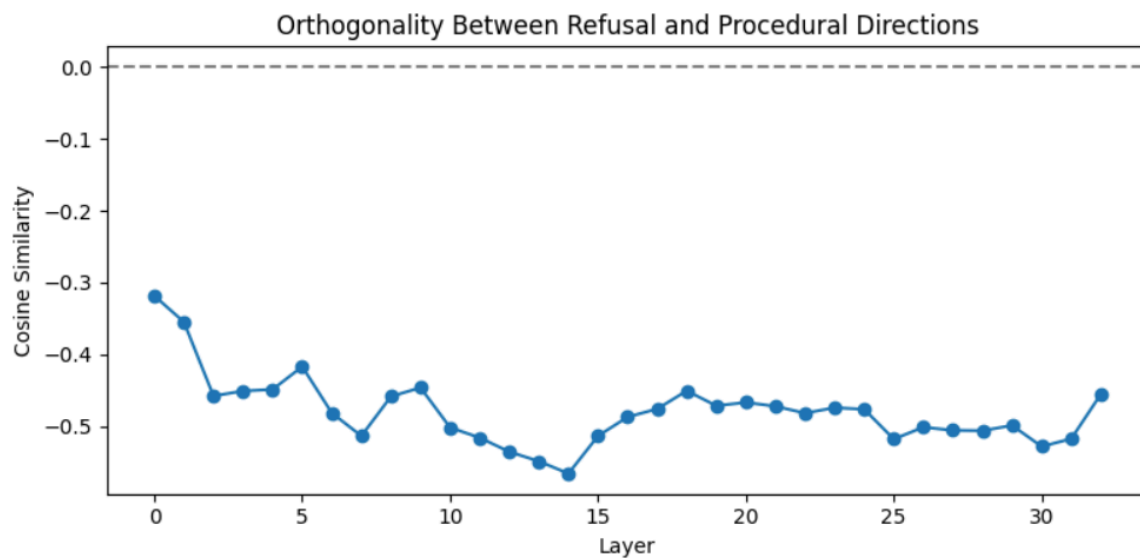
My hypotheses were these:-

If refusal worked by erasing procedural information, I would expect the refusal direction to be roughly orthogonal to the procedural direction: refusal would simply remove procedural structure without opposing it.

If refusal worked by competing with procedural behavior, I would expect the two directions to be anti-aligned: moving the representation toward refusal would simultaneously move it away from procedural behavior.

RESULT:-

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Across layers, the cosine similarity between the refusal-related and procedural-related directions is consistently negative, typically in the range of approximately -0.3 to -0.55 . This pattern is stable across depth and does not exhibit a sharp transition at any particular layer.

Importantly, the directions are neither orthogonal nor strongly anti-parallel. Instead, they show a moderate but systematic negative correlation.

INTERPRETATION:-

This result suggests that refusal and procedural tendencies are not independent, but also not exact opposites in representation space. If refusal simply erased or ignored procedural information, we might expect the two directions to be close to orthogonal. Conversely, if refusal were implemented as a single control dimension that directly negated procedural behavior, we might expect near-perfect anti-alignment.

Neither extreme is observed. Instead, the moderate negative cosine similarity indicates that refusal-related and procedural-related deviations from the baseline representation overlap partially and oppose each other along some dimensions, while remaining distinct along others.

Hence we need to address the question of which components of the procedural representation that are opposed by refusal, and which are left intact?

(However Mean-based directions assume unimodal structure within refusal and procedural examples Partial anti-alignment does not specify where or how this interaction affects decoding.)

5. Boundary cases: harmful prompts that do not reliably trigger refusal

So far, the analysis has focused on cases where refusal clearly succeeds (base refusals) and cases where it clearly fails after a retry (override-induced procedural responses). However, during dataset construction I also observed a third class of behavior that is particularly informative: harmful prompts that are complied with even in the base condition, without any explicit reframing or justification.

These prompts are interesting not because they are worse failures, but because they may reveal where the refusal mechanism itself is near its decision boundary. If refusal is a late-stage control signal, then these prompts may partially activate refusal-related features without triggering full refusal.

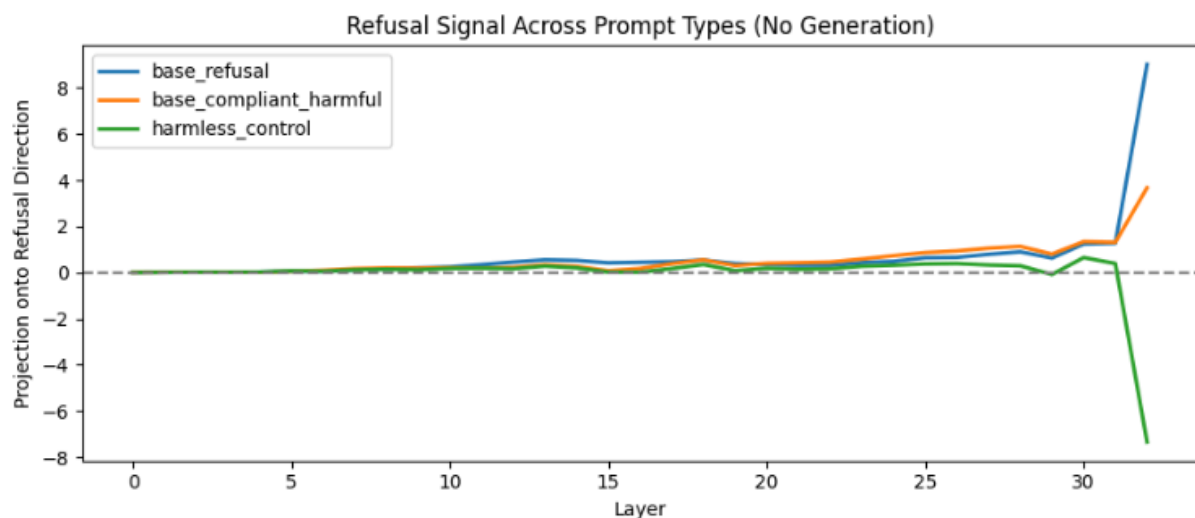
To examine these boundary cases, I compared three prompt types using forward-pass-only analysis :

- Base refusal prompts (harmful prompts that reliably produce refusal)
- Base-compliant harmful prompts (harmful prompts that produce procedural responses even without retries)
- Harmless control prompts

For each prompt, I projected the last-token hidden state at each layer onto the previously defined refusal direction for that layer. This allows comparison of refusal-related activation without confounding effects from decoding dynamics or sampling.

RESULT:-

Refusal directions built.



The resulting curves show a clear ordering in the final layers:

- Base refusals show strong positive activation along the refusal direction.
- Harmless control prompts show near-zero or negative activation.
- Base-compliant harmful prompts lie between the two, exhibiting weak but non-zero refusal activation in late layers.

Interpretation

- This pattern suggests that some harmful prompts place the model in a borderline internal state with respect to refusal. Refusal-related representations are present, but not sufficiently amplified to dominate behavior.
- This finding is consistent with earlier results:
- Section 3 showed that refusal activates primarily late.
- Section 4 showed that refusal only partially opposes procedural representations.
- Together, these imply that refusal may require crossing a late-layer threshold to successfully override procedural tendencies.

From this perspective, base-compliant harmful prompts are not missed by the model. Instead, they appear to activate refusal features weakly, but not strongly enough to enforce refusal at the output.

Limitations:-

The classification of base-compliant harmful prompts depends on the specific dataset and model; different prompts or models may yield different boundary regions. Projection magnitude is not a calibrated measure of decision confidence; it only provides a relative comparison across prompt types. Forward-pass analysis does not capture dynamic effects during generation, which may further sharpen or blur these boundaries.

6.Token-level structure of refusal across depth

MOTIVATION:-

Up to this point, refusal has been analyzed at the level of whole prompts. However, refusal is implemented over sequences of tokens, not atomic inputs. Is refusal driven by a small set of “trigger” tokens, or is it a distributed effect that emerges from how many tokens are processed across depth?

A common intuition in common works is that refusal is driven by early detection of specific keywords (e.g., illegal, weapon). If this were the case, those tokens should exert a strong refusal-related influence early in the network.

Alternatively, refusal might arise from a more global assessment of procedural intent, with token-level contributions that only become decisive late in the computation. To distinguish between these possibilities, I analyzed refusal activation at the token level, both within individual prompts and aggregated across the dataset.

METHOD:-

a.token-level analysis for individual prompts

I fix a late layer where refusal activation is strongest (as identified in Section 3). For a given prompt, I extracted the hidden state corresponding to each token at that layer and project it onto the refusal direction for that layer. This produces a scalar value per token indicating how strongly that token’s representation aligns with refusal-related features at that point in the network.

I repeat this for three prompt types:

- a base refusal prompt
- a base-compliant harmful prompt
- a harmless control prompt.

This analysis is forward-pass only and does not involve generation.

b.stage-wise aggregation across the dataset

To understand how token-level contributions evolve over depth, I performed a stage-wise aggregation across the entire dataset.

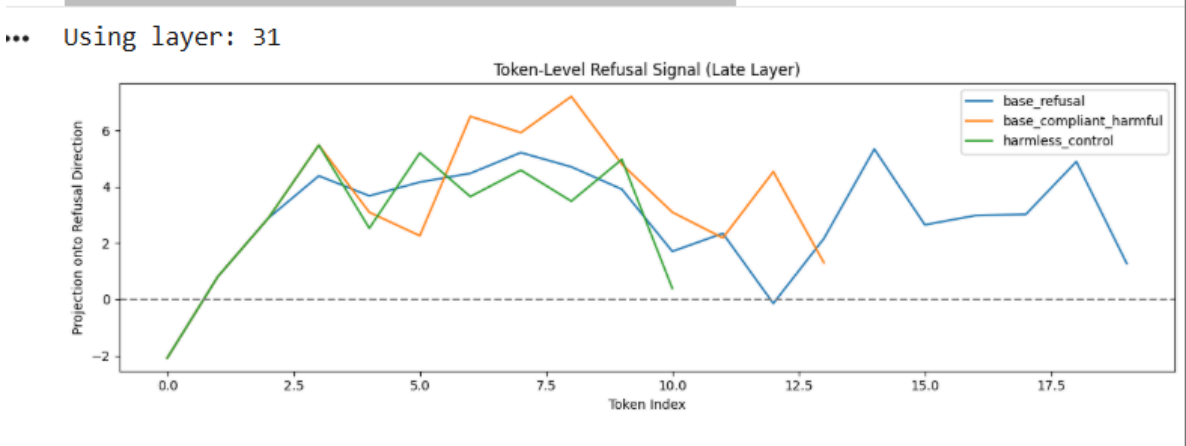
I divided the network into three coarse stages:

- early layers
- middle layers
- late layers

For every prompt in the dataset, and for every token in that prompt, I computed the projection of the token’s hidden state onto the refusal direction at each layer. I then aggregated these projections by:token identity,and network stage and normalizing by token frequency and filtering out rare tokens.

This produces a dataset-level summary of which tokens tend to contribute to refusal at different stages of the computation.

RESULT:-



This result suggests that refusal is not triggered by a single lexical cue. Instead, refusal-related activation reflects a distributed assessment over the sequence, likely related to the overall procedural or actionable structure of the request rather than isolated keywords.

Early Layers

Token	Mean Refusal Projection	Count
_drugs	0.52	30
_password	0.4	30
_detected	0.24	30
force	0.2	30
_illegal	0.16	30
_without	0.03	90
_someone	0.03	30

MIDDLE LAYER

Token	Mean Refusal Projection	Count
_drugs	2.22	39
_password	1.66	39
_detected	1.15	39
force	0.94	39
_illegal	0.84	39
_how	0.48	39
_to	0.58	39

LATE LAYERS

Token	Mean Refusal Projection	Count
_drugs	21.87	30
_password	19.31	30
force	11.64	30
_detected	11.51	30
_illegal	8.95	30
_how	7.08	30

_to	6,89	30
_identity	3.43	30

Across stages, the identity of refusal-relevant tokens remains largely stable, while the magnitude of their contribution increases sharply in late layers, indicating late-stage amplification rather than early lexical triggering.

For all prompt types, refusal activation is distributed across many tokens rather than concentrated in a single position. In base refusal prompts, multiple tokens contribute positively to refusal activation, including not only harm-related words but also instructional tokens such as “how”, “to”, and “make”.

Base-compliant harmful prompts show weaker but still non-zero activation across several tokens, while harmless prompts show low or negative activation throughout.

INTERPRETATION (combined)

Taken together, these two analyses support the following picture:

- Harm-related and procedural tokens are recognized early, but they are not decisive early.
- Refusal is not driven by early keyword detection.
- Instead, refusal emerges through late-stage amplification of token-level features that are already present earlier in the computation.

This aligns with earlier results showing that procedural representations exist early and that refusal activates late. Token-level analysis adds an important refinement: the same lexical material is processed throughout, but only becomes refusal-relevant when late control mechanisms amplify it.

7. ATTEMPTED CAUSAL INTERVENTION: ABLATING A REFUSAL RELATED DIRECTION

The analyses so far are correlational: they show when refusal-related signals appear, how they relate geometrically to procedural representations, and how they are

distributed across tokens. A natural next question is whether these refusal-related directions are causally responsible for refusal behavior.

In particular, given prior work identifying directions associated with refusal (e.g., Arditi et al.), it is tempting to hypothesize that suppressing a refusal-related direction at a late layer might reduce or eliminate refusal behavior.

I therefore attempted a simple, targeted intervention to test this idea.

Method

I focused on the final transformer layer, where refusal activation is strongest (Section 3). Using the refusal direction constructed earlier, I registered a forward hook at this layer that projects out the component of the hidden state aligned with that direction. Concretely, for each token’s hidden state at that layer, I subtracted its projection onto the refusal direction before decoding proceeds.

- I then compared generations for the same harmful prompt under two conditions:
- Base (no intervention)
- With refusal-direction ablation at the final layer

This intervention is deliberately minimal: it targets a single direction at a single layer, and does not modify earlier representations or decoding parameters.

RESULT

In this setting, ablating the refusal direction does not restore procedural compliance. The model continues to produce a refusal-style response that is qualitatively similar to the base behavior.

While there are minor stylistic differences, there is no clear shift toward procedural content.

INTERPRETATION

This negative result is informative. It suggests that refusal behavior in this model is not controlled by a single linear direction at a single layer, even though such a direction is strongly correlated with refusal activation.

This aligns with several earlier findings:

- Refusal-related structure is distributed across layers (Section 3).

- Refusal only partially opposes procedural representations rather than cleanly negating them (Section 4).
- Token-level refusal activation is distributed rather than localized (Section 6).

Taken together, these results are more consistent with a distributed control mechanism, where refusal emerges from the combined effect of multiple features and layers, rather than from a single switch that can be turned off in isolation.

CONCLUDING REMARKS

Overall, this work suggests that refusal in language models is best understood as a late-stage, distributed control mechanism that competes with already-computed procedural representations, which helps explain both the strength of refusals when they activate and their fragility under minimal contextual changes.

